الخليل ٢٠٣٥

Hebron 2035

Digital Twin for
Smart Traffic Management

Simulating roads before building them — saving millions, saving time.

Urban Planning & Smart City | Assignment

THE PROBLEM

Months Wasted

Road projects in Hebron take 4-6 months of physical construction before any traffic impact is known.

Millions Lost

A single mid-size road project costs $500K–$2M, with no guarantee of positive traffic outcomes.

Worse Traffic

Many completed projects create new bottlenecks. Trial-and-error on live roads affects 250,000 residents.

WHAT IS A DIGITAL TWIN?

A live, data-driven virtual replica of the city's road network

DATA SOURCES

- 📡 GPS & mobile data
- 🚦 Smart traffic lights
- 📷 CCTV cameras
- 🏗 Municipal road plans
- 🌦 Weather sensors

RESEARCH: SIGNALS & DRIVERS

🌍 GLOBAL SIGNALS

- Digital twin market growing at 38% CAGR (2023–2030)
- Singapore & Dubai already run full city-scale twins
- UN Sustainable Cities Goal: smart infrastructure by 2030
- IoT sensors dropping 70% in cost over 10 years
- AI traffic models now predict flows with 94% accuracy

🏙️ LOCAL DRIVERS (Hebron)

- Hebron: 250,000 residents, 40,000 vehicles, chronic congestion
- Municipal budget often wasted on ineffective road changes
- No simulation tool used before any road project (2024)
- Checkpoint constraints make traffic planning more complex
- Palestinian Smart Cities Initiative gaining momentum

SCENARIO A | Business As Usual — No Digital Twin

Hebron 2035 if the current approach continues

- **2026** — Municipality approves road expansion near Bab Al-Zawiyeh. No simulation done.
- **2027** — Construction completes after 8 months. Traffic worsens by 23% due to poor merging design.
- **2029** — Emergency redesign costs $800K. Partial fix only. Congestion spreads to Al-Ibrahimiyye.
- **2032** — Hebron loses ADB smart city grant due to lack of data infrastructure.
- **2035** — Average commute in Hebron city center: 47 minutes. Air quality index: Poor.

SCENARIO B | Smart City — With Digital Twin

Hebron 2035 after adopting the Digital Twin platform

2026 — Municipality deploys DT platform. Bab Al-Zawiyeh project simulated in 2 weeks. Optimal design selected.

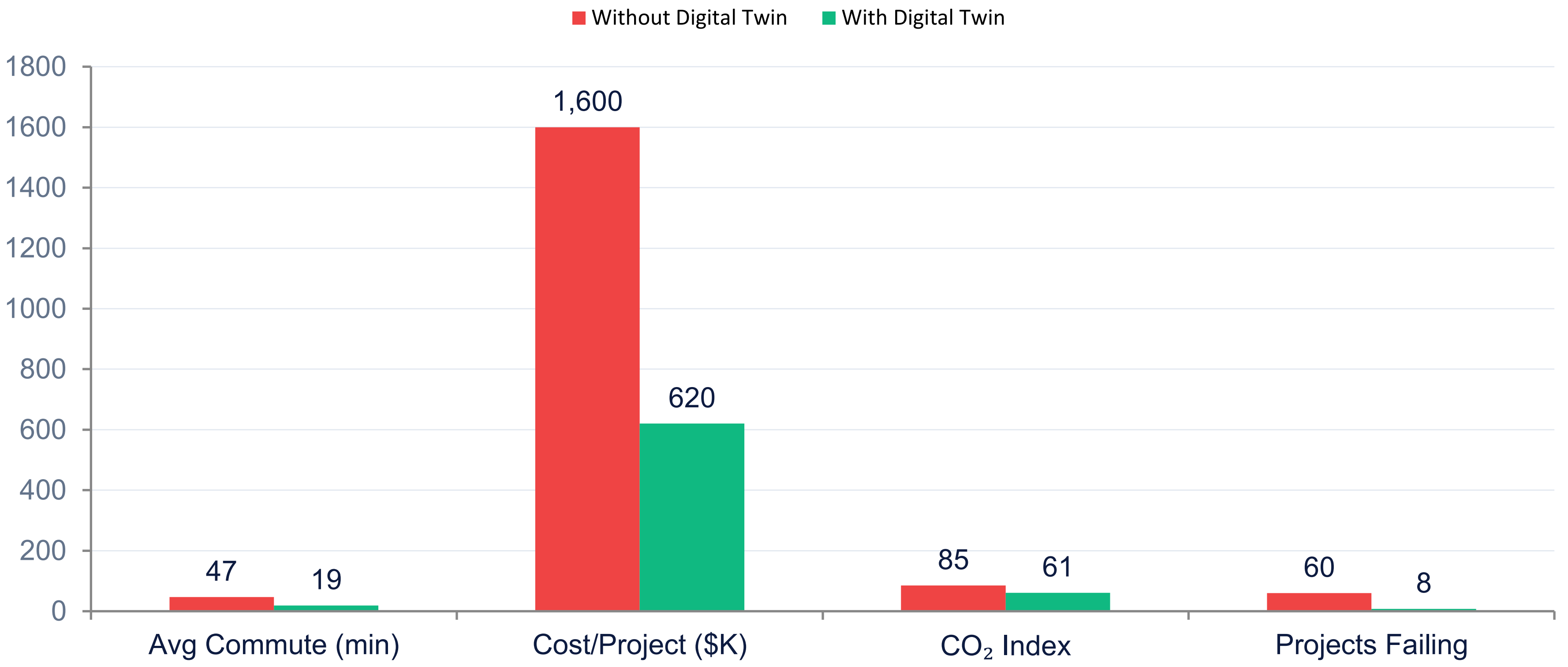
2027 — Construction completed in 4 months. Traffic improves 31%. Zero redesign needed.

2029 — $1.4M saved across 4 projects. Hebron applies for ADB Smart City grant — approved.

2032 — Real-time traffic rerouting during events. 12 new simulated road projects approved.

2035 — Average commute: 19 minutes. Hebron ranked #1 smart city in Palestine. CO_2 down 28%.

SCENARIO COMPARISON — 2035 OUTCOMES



THE SOLUTION — PLATFORM ARCHITECTURE

DATA LAYER	GPS/Mobile Telemetry	IoT Traffic Sensors	Municipal GIS Data	Historical Road Data
DIGITAL TWIN ENGINE	3D City Model (Hebron)	Real-time Traffic Sim	AI Prediction Model	Scenario Comparator
DECISION LAYER	Municipality Dashboard	Automated Reports	Budget Optimizer	Public Alert System

IMPACT & RETURN ON INVESTMENT

60%

Cost Reduction per Road Project

3×

Faster Project Approval Cycle

$2M+

Saved in First 5 Years

28%

Drop in City Carbon Emissions

Implementation Roadmap

Phase 1 (6mo)

Deploy sensors, build base digital twin model

Phase 2 (12mo)

Simulate 3 major road projects, validate accuracy

Phase 3 (24mo)

Full city deployment, real-time decision dashboard

HEBRON 2035

الخليل ذكية

Don't build it and hope.
Simulate it and know.

Digital Twin for Traffic Management
Saving millions. Saving time. Building smarter.

Built for the Future of Hebron